

HAT-P-37b

R-2663

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_160919 · created 2026-08-30T00:35:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457650.7038 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.044
Transit depth (Rp/Rs)^2	1.51 +/- 1.08 [%]
Orbital Inclination (inc)	84.86 +/- 2.2
Airmass coefficient 1 (a1)	1.017 +/- 0.026
Airmass coefficient 2 (a2)	-0.012 +/- 0.011
Scatter in the residuals of the lightcurve fit is	2.57 %
Best Comparison Star	#3 - [510, 216]
Optimal Aperture	2.54
Optimal Annulus	8.22
Transit Duration (day)	0.063 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.044 vs 0.1394 (deviation 0.37σ)
Duration fit vs published	0.063 d vs 0.0983 d (deviation 1.04σ)
Mid-time O–C	-4.7 min
Depth SNR	2.8
Residual scatter	2.57 %

Light curve

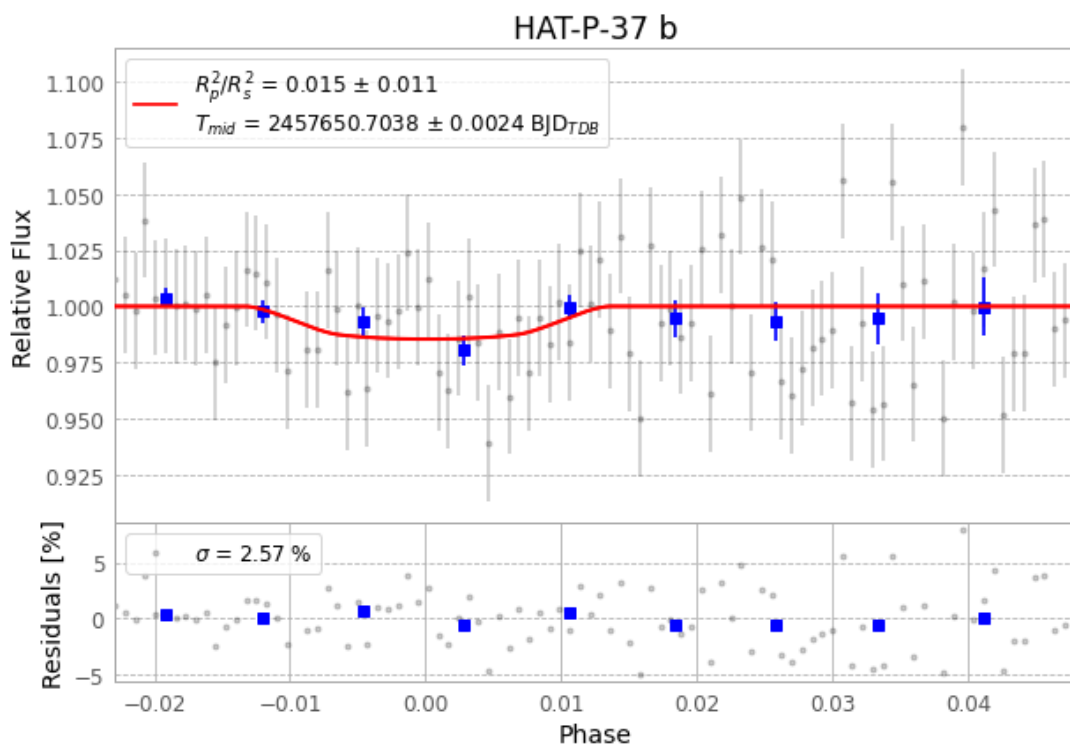


Figure 1 — FinalLightCurve_HAT-P-37 b_2016-09-18.png (SHA-256 f4d8113c37c69c08...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [516, 259], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 71e4d7bca60faa2a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

96 FITS frames (63 MB), HATP-37160919031812.FITS → HATP-37160919080312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-160919.log (a2002b4397cc...), FinalLightCurve_HAT-P-37 b_2016-09-18.png (f4d8113c37c6...), FinalLightCurve_HAT-P-37 b_2016-09-18.csv (daba32d7a2dc...)

Compute resources

Wall time 133.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:35Z.